VORTEX-HRM

Hierarchical Retrieval-Augmented Generation for Multi-Hop Question Answering

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Problem

Multi-hop QA needs multiple reasoning + retrieval steps. Standard RAG retrieves once—single-hop only.

`figs/architecture.png`

Design Principles

- **Fact-Free Synapses** — 0% facts in planner weights, 100% routing
- **Entropic Collapse** — convergence via entropy plateau
- **Offline-First** — zero API keys, any hardware

Gravitational Core (Planner)

Holds zero factual knowledge. State: goal_vector, spiral_memory, hop_history, confidence/entropy. Output: XML `<step>`, `<final_answer>`, `<stop_search>`.

Centrifugal Ingestor (Executor)

Fixed chained pipeline (no LLM tool selection):

1. keyword_search → lexical match
2. chunk_read → full text + neighbors
3. LLM → condensed `<fact>`

Cyclic Spiral Flow

Planner → `<step>` → Executor → `<fact>` → … → convergence.

Termination: confidence $\geq$ 0.85, entropy stall ×3, or max hops (15).

Benchmark

Setup: 50 multi-domain QA, 60 chunks, qwen2.5:7b (CPU), ×2 runs

Metric	Avg	Note
Contains	**71%**	Truth in prediction
Token F1	22.5%	Partial overlap
Exact Match	0%	Format issue
Avg Spirals	1.3	64% in 0–1

Key Findings

- Architecture validated: 72% Contains on 7B CPU
- Model size critical: $\leq$0.5B fails XML contract
- Fixed pipeline beats LLM tool selection
- EM 40%+ expected with GPT-4o-mini (Phase 2)

Next Steps

- Full HotpotQA evaluation (500 questions)
- Semantic search (FAISS)
- GPT-4o-mini backend
- Comparison with A-RAG baselines